METRIC

MIL-DTL-83133K
18 July 2018

SUPERSEDING
MIL-DTL-83133J
16 December 2015

DETAIL SPECIFICATION

TURBINE FUEL, AVIATION, KEROSENE TYPE, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three grades of kerosene type aviation turbine fuel, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37). Synthesized hydrocarbons from new sources require specific guidance that is outside the scope of MIL-DTL-83133. This guidance is found in ASTM D7566.

1.2 Classification. Aviation turbine fuel will be of the following grades, as specified (see 6.2).

Grade	NATO Code No.	Description
JP-8	F-34	Kerosene-type turbine fuel which will contain a static dissipater additive, corrosion inhibitor/lubricity improver, and fuel system icing inhibitor, and may contain antioxidant and metal deactivator additives.
---	F-35	Kerosene-type turbine fuel which will contain a static dissipater additive, may contain antioxidant, corrosion inhibitor/lubricity improver, and metal deactivator additives but will not contain fuel system icing inhibitor.
JP-8+100	F-37	JP-8 type kerosene turbine fuel which contains thermal stability improver additive (NATO S-1749) as described in 3.3.6.

Comments, suggestions, or questions on this document should be addressed to AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB OH 45433-7631 or e-mailed to AFPET.PTPS@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-85470	-	Inhibitor, Icing, Fuel System, High Flash NATO Code Number S-1745
MIL-PRF-25017	-	Inhibitor, Corrosion/Lubricity Improver, Fuel Soluble (NATO S-1747)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-290	-	Packaging and Marking of Petroleum and Related Products

QUALIFIED PRODUCTS LIST

QPL-25017	-	Inhibitor, Corrosion/Lubricity Improver, Fuel Soluble (NATO S-1747)

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D56	-	Standard Test Method for Flash Point by Tag Closed Cup Tester
ASTM D86	-	Standard Test Method for Distillation of Petroleum Products and Liquid Fuels at Atmospheric Pressure
ASTM D93	-	Standard Test Methods for Flash Point by Pensky-Martens Closed Cup Tester

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ASTM D129	-	Standard Test Method for Sulfur in Petroleum Products (General High Pressure Decomposition Device Method)
ASTM D130	-	Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test
ASTM D156	-	Standard Test Method for Saybolt Color of Petroleum Products (Saybolt Chromometer Method)
ASTM D381	-	Standard Test Method for Gum Content in Fuels by Jet Evaporation
ASTM D445	-	Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)
ASTM D976	-	Standard Test Method for Calculated Cetane Index of Distillate Fuels
ASTM D1266	-	Standard Test Method for Sulfur in Petroleum Products (Lamp Method)
ASTM D1298	-	Standard Test Method for Density, Relative Density, or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method
ASTM D1319	-	Standard Test Method for Hydrocarbon Types in Liquid Petroleum Products by Fluorescent Indicator Adsorption
ASTM D1322	-	Standard Test Method for Smoke Point of Kerosene and Aviation Turbine Fuel
ASTM D1840	-	Standard Test Method for Naphthalene Hydrocarbons in Aviation Turbine Fuels by Ultraviolet Spectrophotometry
ASTM D2276	-	Standard Test Method for Particulate Contaminant in Aviation Fuel by Line Sampling
ASTM D2386	-	Standard Test Method for Freezing Point of Aviation Fuels
ASTM D2622	-	Standard Test Method for Sulfur in Petroleum Products by Wavelength Dispersive X-Ray Fluorescence Spectrometry
ASTM D2624	-	Standard Test Methods for Electrical Conductivity of Aviation and Distillate Fuels
ASTM D2887	-	Standard Test Method for Boiling Range Distribution of Petroleum Fractions by Gas Chromatography
ASTM D3120	-	Standard Test Method for Trace Quantities of Sulfur in Light Liquid Petroleum Hydrocarbons by Oxidative Microcoulometry

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ASTM D3227 - Standard Test Method for (Thiol Mercaptan) Sulfur in Gasoline, Kerosine, Aviation Turbine, and Distillate Fuels (Potentiometric Method)

ASTM D3241 - Standard Test Method for Thermal Oxidation Stability of Aviation Turbine Fuels

ASTM D3242 - Standard Test Method for Acidity in Aviation Turbine Fuel

ASTM D3338 - Standard Test Method for Estimation of Net Heat of Combustion of Aviation Fuels

ASTM D3343 - Standard Test Method for Estimation of Hydrogen Content of Aviation Fuels

ASTM D3701 - Standard Test Method for Hydrogen Content of Aviation Turbine Fuels by Low Resolution Nuclear Magnetic Resonance Spectrometry

ASTM D3828 - Standard Test Methods for Flash Point by Small Scale Closed Cup Tester

ASTM D3948 - Standard Test Method for Determining Water Separation Characteristics of Aviation Turbine Fuels by Portable Separometer

ASTM D4052 - Standard Test Method for Density, Relative Density, and API Gravity of Liquids by Digital Density Meter

ASTM D4057 - Standard Practice for Manual Sampling of Petroleum and Petroleum Products

ASTM D4177 - Standard Practice for Automatic Sampling of Petroleum and Petroleum Products

ASTM D4294 - Standard Test Method for Sulfur in Petroleum and Petroleum Products by Energy-Dispersive X-Ray Fluorescence Spectrometry

ASTM D4306 - Standard Practice for Aviation Fuel Sample Containers for Tests Affected by Trace Contamination

ASTM D4529 - Standard Test Method for Estimation of Net Heat of Combustion of Aviation Fuels

ASTM D4737 - Standard Test Method for Calculated Cetane Index by Four Variable Equation

ASTM D4809 - Standard Test Method for Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter (Precision Method)

ASTM D4952 - Standard Test Method for Qualitative Analysis for Active Sulfur Species in Fuels and Solvents (Doctor Test)

ASTM D5006 - Standard Test Method for Measurement of Fuel System Icing Inhibitors (Ether Type) in Aviation Fuels

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ASTM D5291	-	Standard Test Methods for Instrumental Determination of Carbon, Hydrogen, and Nitrogen in Petroleum Products and Lubricants
ASTM D5452	-	Standard Test Method for Particulate Contamination in Aviation Fuels by Laboratory Filtration
ASTM D5453	-	Standard Test Method for Determination of Total Sulfur in Light Hydrocarbons, Spark Ignition Engine Fuel, Diesel Engine Fuel, and Engine Oil by Ultraviolet Fluorescence
ASTM D5972	-	Standard Test Method for Freezing Point of Aviation Fuels (Automatic Phase Transition Method)
ASTM D6045	-	Standard Test Method for Color of Petroleum Products by the Automatic Tristimulus Method
ASTM D6379	-	Standard Test Method for Determination of Aromatic Hydrocarbon Types in Aviation Fuels and Petroleum Distillates—High Performance Liquid Chromatography Method with Refractive Index Detection
ASTM D6751	-	Standard Specification for Biodiesel Fuel Blend Stock (B100) for Middle Distillate Fuels
ASTM D6890	-	Standard Test Method for Determination of Ignition Delay and Derived Cetane Number (DCN) of Diesel Fuel Oils by Combustion in a Constant Volume Chamber
ASTM D6986	-	Standard Test Method for Free Water, Particulate and Other Contamination in Aviation Fuels (Visual Procedure Procedures)
ASTM D7042	-	Standard Test Method for Dynamic Viscosity and Density of Liquids by Stabinger Viscometer (and the Calculation of Kinematic Viscosity)
ASTM D7153	-	Standard Test Method for Freezing Point of Aviation Fuels (Automatic Laser Method)
ASTM D7154	-	Standard Test Method for Freezing Point of Aviation Fuels (Automatic Fiber Optical Method)
ASTM D7170	-	Standard Test Method for Determination of Derived Cetane Number (DCN) of Diesel Fuel Oils—Fixed Range Injection Period, Constant Volume Combustion Chamber Method
ASTM D7171	-	Standard Test Method for Hydrogen Content of Middle Distillate Petroleum Products by Low-Resolution Nuclear Magnetic Resonance Spectroscopy
ASTM D7224	-	Standard Test Method for Determining Water Separation Characteristics of Kerosine-Type Aviation Turbine Fuels Containing Additives by Portable Separometer

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ASTM D7345	-	Standard Test Method for Distillation of Petroleum Products and Liquid Fuels at Atmospheric Pressure (Micro Distillation Method)
ASTM D7566	-	Standard Specification for Aviation Turbine Fuel Containing Synthesized Hydrocarbons
ASTM D7619	-	Standard Test Method for Sizing and Counting Particles in Light and Middle Distillate Fuels, by Automatic Particle Counter
ASTM D7777	-	Standard Test Method for Density, Relative Density, or API Gravity of Liquid Petroleum by Portable Digital Density Meter
ASTM D7797	-	Standard Test Method for Determination of the Fatty Acid Methyl Esters Content of Aviation Turbine Fuel Using Flow Analysis by Fourier Transform Infrared Spectroscopy – Rapid Screening Method
ASTM D7945	-	Standard Test Method for Determination of Dynamic Viscosity and Derived Kinematic Viscosity of Liquids by Constant Pressure Viscometer
ASTM D8166	-	Standard Test Method for Sizing and Counting Particulates in Light and Middle Distillate Fuels and Biodiesel Blend (B6 to B20) Using Continuous Flow and Bottle Sampler Automatic Particle Counters
ASTM E29	-	Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications

(Copies of these documents are available from https://www.astm.org.)

ENERGY INSTITUTE

IP 12	-	Determination of specific energy
IP 16	-	Determination of the freezing point of aviation fuels - Manual method
IP 71 (Section 1)	-	Petroleum products - Transparent and opaque liquids - Determination of kinematic viscosity and calculation of dynamic viscosity
IP 123	-	Petroleum products - Determination of distillation characteristics at atmospheric pressure
IP 154	-	Petroleum products - Corrosiveness to copper - Copper strip test
IP 156	-	Petroleum products and related materials - Determination of hydrocarbon types - Fluorescent indicator adsorption method

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IP 160	-	Crude petroleum and liquid petroleum products - Laboratory determination of density - Hydrometer method
IP 170	-	Determination of flash point - Abel closed-cup method
IP 216	-	Determination of particulate contaminant of aviation turbine fuels by line sampling
IP 274	-	Determination of electrical conductivity of aviation and distillate fuels
IP 323	-	Determination of thermal oxidation stability of gas turbine fuels
IP 336	-	Petroleum products - Determination of sulfur content - Energy-dispersive-X-ray fluorescence spectrometry
IP 342	-	Petroleum products - Determination of thiol (mercaptan) sulfur in light and middle distillate fuels - Potentiometric method
IP 354	-	Determination of the acid number of aviation turbine fuels - Colour-indicator titration method
IP 365	-	Crude petroleum and petroleum products - Determination of density - Oscillating U-tube method
IP 406	-	Petroleum products - Determination of boiling range distribution by gas chromatography
IP 423	-	Determination of particulate contaminant in aviation turbine fuels by laboratory filtration
IP 424	-	Determination of fuel system icing inhibitor content of aviation turbine kerosines by high performance liquid chromatography
IP 435	-	Determination of the freezing point of aviation turbine fuels by the automatic phase transition method
IP 436	-	Determination of aromatic hydrocarbon types in aviation fuels and petroleum distillates - High performance liquid chromatography method with refractive index detection
IP 528	-	Determination of the freezing point of aviation turbine fuels - Automated fibre optic method
IP 529	-	Determination of the freezing point of aviation fuels - Automatic laser method
IP 540	-	Determination of the existent gum content of aviation turbine fuel - Jet evaporation method
IP 564	-	Determination of the level of cleanliness of aviation turbine fuel - Laboratory automatic particle counter method

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IP 565	-	Determination of the level of cleanliness of aviation turbine fuel - Portable automatic particle counter method
IP 577	-	Determination of the level of cleanliness of aviation turbine fuel - Automatic particle counter method using light extinction
IP 583	-	Determination of the fatty acid methyl esters content of aviation turbine fuel using flow analysis by Fourier transform infrared spectroscopy - Rapid screening method
IP 585	-	Determination of fatty acid methyl esters (FAME), derived from bio-diesel fuel, in aviation turbine fuel - GC-MS with selective ion monitoring/scan detection method
IP 590	-	Determination of fatty acid methyl esters (FAME) in aviation turbine fuel - HPLC evaporative light scattering detector method
IP 598	-	Petroleum Products - Determination of the smoke point of kerosine, manual and automated method
IP 599	-	Determination of Fatty Acid Methyl Esters (FAME) in aviation turbine fuel - Gas Chromatography using heart-cut and refocusing

(Copies of these documents are available from https://publishing.energyinst.org/.)

EUROPEAN COMMITTEE FOR STANDARDIZATION

EN 14214	-	Liquid petroleum products - Fatty acid methyl esters (FAME) for use in diesel engines and heating applications - Requirements and test methods

(Copies of this document are available from https://www.cen.eu.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 4406	-	Hydraulic fluid power - Fluids - Method for coding the level of contamination by solid particles
ISO 11171	-	Hydraulic fluid power - Calibration of automatic particle counters for liquids

(Copies of these documents are available from https://www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. Unless otherwise specified (see 3.1.1), fuel supplied under this specification shall be refined hydrocarbon distillate fuel oil containing additives in accordance with 3.3. The feedstock from which the fuel is refined shall be crude oils derived from petroleum, oil sands, oil shale, or mixtures thereof.

3.1.1 Synthesized materials. Up to 50 volume percent of the finished fuel may consist of Synthesized Paraffinic Kerosene (SPK) derived from a Fischer-Tropsch (FT) process meeting the requirements of ASTM D7566 Annex A1 (Fischer-Tropsch Hydroprocessed Synthesized Paraffinic Kerosine) or SPKs derived from Hydroprocessed Esters and Fatty Acids (HEFA) meeting the requirements of ASTM D7566 Annex A2 (Synthesized Paraffinic Kerosine from Hydroprocessed Esters and Fatty Acids). HEFA SPK has also been called Hydroprocessed Renewable Jet or Hydrotreated Renewable Jet (HRJ) and, for the purpose of this specification, the terms are considered interchangeable.

3.2 Finished fuel. Finished fuels shall meet the requirements of this specification. The requirements of Table III apply if the finished fuel contains synthesized materials.

3.3 Additives. The type and amount of each additive used shall be made available when requested by procuring activity or user (see 6.2.d). The only additives approved for use are those referenced in this specification.

3.3.1 Antioxidants. Immediately after processing and before the fuel is exposed to the atmosphere (such as during rundown into feed/batch tankage), add an approved antioxidant formulation (3.3.1.1) or combination of approved antioxidant formulations in order to prevent the formation of gums and peroxides after manufacture. The concentration of antioxidant to be added shall be:

a. Not less than 17.2 milligrams (mg) nor more than 24.0 mg of active ingredient per liter (L) of fuel (6.0 to 8.4 lb/1000 barrels) to all JP-8 fuel that contains blending stocks that have been hydrogen treated or SPK derived from hydrotreated, hydrocracked, or hydroisomerized products of a Fischer-Tropsch or HEFA process.

b. At the option of the supplier, not more than 24.0 mg of active ingredient per liter of fuel (8.4 lb/1000 barrels) may be added to JP-8 fuels that do not contain hydrogen treated blending stocks or SPK derived from hydrotreated, hydrocracked, or hydroisomerized products of a Fischer-Tropsch or HEFA process.

3.3.1.1 Antioxidant formulations. The following antioxidant formulations are approved:

a. 2,6-ditertiary-butyl-4-methyl-phenol

b. 2,4-dimethyl-6-tertiary-butyl-phenol

c. 2,6-di-tertiary-butyl-phenol

d. 75 percent minimum 2,6-di-tertertiary-butyl-phenol
25 percent maximum tertiary and tritertiarybutyl-phenols

e. 72 percent minimum 6-tertiary-butyl-2,4-dimethyl-phenol
28 percent maximum mixture of tertiary-butyl-methyl-phenols and tertiary-butyl-dimethyl-phenols

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f. 55 percent minimum 2,4-dimethyl-6-tertiary-butyl-phenol, and
15 percent minimum 2,6-di-tertiary -4-methyl-phenol, and
30 percent maximum mixed methyl and dimethyl tertiary-butyl-phenols

3.3.2 Metal deactivator. Metal deactivator additive, N,N'-disalicylidene-1,2-propane diamine, shall not be used in JP-8 unless the supplier has obtained written consent from the procuring activity and user. The concentration of active material used on initial batching of the fuel at the refinery shall not exceed 2.0 mg/L. Cumulative addition of metal deactivator when re-doping the fuel shall not exceed 5.7 mg/L.

3.3.3 Static dissipater additive. An additive shall be blended into the fuel in sufficient concentration to increase the conductivity of the fuel to within the range specified in Table I at the point of injection. The point of injection of the additive shall be determined by agreement between the purchasing authority and the supplier. The following electrical conductivity additives are approved: Stadis® 450 manufactured by Innospec Fuel Specialties, LLC and AvGuard® SDA manufactured by Afton Chemical Corporation.

3.3.4 Corrosion inhibitor/lubricity improver additive. A corrosion inhibitor/lubricity improver (CI/LI) additive meeting the requirements of MIL-PRF-25017 and approved in Qualified Products List (QPL)-25017 shall be blended into JP-8 (NATO F-34) by the contractor. The CI/LI additive is optional for NATO F-35, unless stated in the contract. The amount added shall be equal to or greater than the minimum effective concentration and shall not exceed the maximum allowable concentration listed in QPL-25017. The contractor or transporting agency, or both, shall maintain and upon request shall make available to the Government evidence that the CI/LI additives used are approved in QPL-25017. The point of injection of the CI/LI additive shall be determined by agreement between the purchasing authority and the supplier.

3.3.5 Fuel system icing inhibitor. The use of a fuel system icing inhibitor meeting the requirements of MIL-DTL-85470 shall be mandatory for JP-8. The point of injection of the additive for JP-8 shall be determined by agreement between the purchasing authority and the supplier. The fuel system icing inhibitor is not to be added to NATO F-35 unless so directed by the purchasing authority.

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TABLE I. Chemical and physical requirements and test methods

Property	Minimum	Maximum	ASTM or IP Test Method
WORKMANSHIP [1]			Section 3.5
COMPOSITION			
Color, Saybolt [1]			D156 [2] or D6045
Total acid number, mg KOH/g		0.015	D3242 / IP 354
Aromatics, vol. percent [3]		25.0	D1319 [2] or IP 156
		26.5	D6379 / IP 436
Sulfur, total, mass percent		0.30	D129, D1266, D2622, D3120 [4], D4294 [2], D5453, or IP 336
Sulfur, mercaptan, mass percent or Doctor test [5]		0.002 negative	D3227 [2] / IP 342 D4952
VOLATILITY			
Distillation temperature, °C [6]			D86 [2,7], D2887 / IP 406, IP 123, or D7345 [8]
Initial boiling point [1]			
10 percent recovered		205	
20 percent recovered [1]			
50 percent recovered [1]			
90 percent recovered [1]			
Final boiling point		300	
Residue, vol. percent		1.5	
Loss, vol. percent		1.5	
Flash point, °C [9]	38		D56, D93 [2], D3828, or IP 170
Density Density, kg/L at 15 °C or Gravity, API at 60 °F	 0.775 37.0	 0.840 51.0	D1298 / IP 160, D4052 [2]/ IP 365, or D7777
FLUIDITY			
Freezing point, °C		-47	D2386 [2] / IP 16, D5972 / IP 435, D7153 / IP 529, or D7154 / IP 528
Viscosity, at -20 °C, mm^2/s		8.0	D445 [2] / IP 71 Section 1, D7042 [10], or D7945
COMBUSTION			
Net heat of combustion, MJ/kg	42.8		D3338, D4529, or D4809 [2] or IP 12
Hydrogen content, mass percent	13.4		D3343, D3701, D5291, or D7171 [2]

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TABLE I. Chemical and physical requirements and test methods – Continued

Property	Minimum	Maximum	Test Method
Smoke point, mm or Smoke point, mm, and Naphthalenes, vol. percent	25.0 19.0	 3.0	D1322 / IP 598 D1322 / IP 598 D1840
Calculated cetane index [1]			D976 or D4737
CORROSION			
Copper strip corrosion, 2 hr at 100 °C (212 °F)		No. 1	D130/IP 154
THERMAL STABILITY			
Thermal stability (2.5 hr at 260 °C) [11]			D3241 / IP 323
Change in pressure drop, mm Hg		25	
Tube rating: One of the following requirements shall be met:			
(1) Annex A1 VTR or		<3 [12]	
(2) Annex A3 ETR or Annex A2 ITR, average deposit thickness, nm, over area of 2.5 mm^2		85	
ADDITIVES			
Fuel system icing inhibitor, vol. percent	0.07	0.10	D5006 [2] or IP 424
Fuel electrical conductivity, pS/m [13]			D2624 / IP 274
CONTAMINANTS			
Existent gum, mg/100 mL [14]		7	D381 [2] or IP 540
Microseparometer Rating [15]			D3948 or D7224 [2]

NOTES:

1. Report.
2. Referee Test Method.
3. Aromatics content shall be determined either by ASTM D1319 (or IP 156) or by ASTM D6379 / IP 436. In case of dispute, ASTM D1319 shall be the referee test method.
4. The sulfur content detection range for ASTM D3120 is 3.0 mg/kg to 1000 mg/kg.
5. If the Doctor Test results in a failure ('positive' result), then mercaptan sulfur content shall be determined by the referee test method ASTM D3227.
6. Distillation property criteria are specified in ASTM D86 scale units. ASTM D2887 results shall be converted to estimated D86 results by application of the correlation in Appendix X4 "Correlation for Jet and Diesel Fuel (Procedures A and B)" of D2887 for comparison with the specified property criteria. Distillation residue and loss limits provide control of the distillation process during the D86 test method and do not apply to D2887.

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TABLE I. Chemical and physical requirements and test methods – Continued

NOTES:

7. ASTM D86 distillation is run at Group 4 conditions except that Group 3 condenser temperatures are used.
8. ASTM D7345 results shall be corrected to bias-free distillation results by the application of the correction factors in the D7345 Precision and Bias section.
9. ASTM D56 may give results up to 1 °C (2 °F) below the ASTM D93 results. ASTM D3828 may give results up to 1.7 °C (3 °F) below the ASTM D93 results. IP 170 is also permitted; may give results up to 2.2 °C (4 °F) below the ASTM D93 results.
10. ASTM D7042 results shall be corrected to bias-free viscosity results by the application of the correction factor in the D7042 Precision and Bias section.
11. See 4.5.3. ASTM D3241 Annex A3 Ellipsometric Tube Rater (ETR) is the referee thermal stability method, when available; otherwise, Annex A2 Interferometric Tube Rater (ITR), when available. Tube deposit failures by ETR or ITR shall be reported as ">85 nm." If the ITR reports "N/A" for the tube's volume measurement, the result constitutes a failure and the result shall be reported as ">85 nm." If test results by either ETR or ITR are reported, then results by D3241 Annex A1 Visual Tube Rater (VTR) are not required.
12. Peacock or Abnormal color deposits result in a failure.
13. The conductivity under the conditions at point of delivery shall be as follows:
    (a) between 150 pS/m and 600 pS/m for JP-8 (NATO F-34)
    (b) between 50 pS/m and 600 pS/m for NATO F-35
    (c) between 150 pS/m to 500 pS/m for JP-8+100 (NATO F-37).
14. The preferred vaporizing medium for aviation turbine fuel is steam; however, the existent gum test IP 540 may be performed using air as the vaporizing medium. If air is used instead of steam, it shall be recorded. Test Method ASTM D381, using steam jet operating conditions, shall be the referee test method.

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TABLE I. Chemical and physical requirements and test methods – Continued

NOTES:

15. The determination of which Micro-Separometer (MSEP) test method is to be run – D3948 or D7224 – will depend on which fuel additives are in the jet fuel. If the jet fuel is fully additized with all the approved fuel additives including SDA, then ASTM D7224 is to be run; otherwise, D3948 is to be run.

(a) The minimum microseparometer rating at point of manufacture using an ASTM D3948 MSEP shall be as follows:

JP-8 Additives	D3948 MSEP Rating, minimum
Antioxidant (AO)*, Metal Deactivator (MDA)*	90
AO*, MDA*, and Fuel System Icing Inhibitor (FSII)	85
AO*, MDA*, and Corrosion Inhibitor/Lubricity Improver (CI/LI)	80
AO*, MDA*, FSII, and CI/LI	70

*Even though the presence or absence of AO and MDA do not change these limits, samples submitted for specification or conformance testing shall contain the same additives present in the refinery batch. Regardless of which minimum the refiner selects to meet, the refiner shall report the MSEP rating on a laboratory hand blend of the fuel with all additives required by the specification.

(b) The MSEP rating requirement by ASTM D7224 is minimum 70. This rating solely applies to a fully additized JP-8 (containing FSII, CI/LI, and SDA; the fuel may also contain AO and MDA).

JP-8 Additives	D7224 MSEP Rating, minimum
FSII, CI/LI, and SDA (may contain AO, MDA)	70

3.3.6 Thermal stability improver additive. Due to logistic concerns, personnel at the operating location shall request written approval from the cognizant activity to add a thermal stability improver additive to the fuel. If approval is given, the concentration of the additive and location of injection shall be specified by the cognizant service activity listed below. For US Air Force aircraft, this approval does not override the single manager's authority for specifying allowed/disallowed fuels. JP-8 fuel with an approved thermal stability improver additive at the required concentration shall be designated as JP-8+100 (NATO F-37). Thermal stability improver additive shall not be used in JP-8 without approval, in writing, from:

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Cognizant activities for the US Army:

a. US Army Ground: RDECOM-TARDEC, Ground System Performance Fluids Team, RDTA-SIE-ES-FPT-GSPF, Building 210, 6501 E. 11 Mile Road, Warren, MI 48397-5000.
b. US Army Aviation: US Army RDECOM, Attn: RDMR-AEP, Building 4488, Room C-211, Redstone Arsenal, AL 35898-5000.

Cognizant activity for the US Navy and US Marine Corps: Naval Air Systems Command, AIR-4.4, Building 2360, 22229 Elmer Road, Patuxent River, MD 20670-1534.

Cognizant activity for the US Air Force: Air Force Petroleum Office, AFPET/PTP, 2430 C Street, Building 70, Area B, Wright-Patterson AFB OH 45433-7631.

3.3.6.1 Qualified additives. Qualified thermal stability improver additives are listed in Table II.

TABLE II. Qualified thermal stability improver additives

Additive Name	Qualification Reference	Manufacturer	Manufacturer Recommended Dosage Rate
SPEC AID 8Q462	AFRL/PRSF Ltr, 9 Dec 1997	Suez Water Technologies & Solutions 9669 Grogan Mill Road The Woodlands, TX 77380	256 mg/L
SPEC AID 8Q462W	ASC/ENFA Tech Eval, 12 Apr 2011	Suez Water Technologies & Solutions 9669 Grogan Mill Road The Woodlands, TX 77380	256 mg/L

3.3.7 Premixing of additives. Additives shall not be premixed with other additives before injection into the fuel so as to prevent possible reactions among the concentrated forms of different additives.

3.4 Additional requirements for finished fuels containing synthesized hydrocarbons. Finished fuels containing synthesized hydrocarbons shall meet the additional requirements specified in Table III, when tested in accordance with the specified test methods.

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TABLE III. Additional requirements of JP-8 containing synthesized hydrocarbons

Property	Minimum	Maximum	Test Method
COMPOSITION			
Aromatics, vol. percent [1]	8.0		D1319 [2] or IP 156
	8.4		D6379 / IP 436
VOLATILITY			
Distillation °C [3]			D86 [2,4] or D2887 / IP 406, or IP 123
50 percent recovery gradient [5]	15		
90 percent recovery gradient [6]	40		
COMBUSTION			
Derived cetane number	40		D6890 [2] or D7170

NOTES:

1. Aromatics content shall be determined either by ASTM D1319 (or IP 156) or by ASTM D6379 / IP 436. In case of dispute, ASTM D1319 shall be the referee test method.
2. Referee Test Method.
3. Distillation property criteria are specified in ASTM D86 scale units. ASTM D2887 results shall be converted to estimated D86 results by application of the correlation in Appendix X4 "Correlation for Jet and Diesel Fuel (Procedures A and B)" of D2887 for comparison with the specified property criteria. Distillation residue and loss limits provide control of the distillation process during the D86 test method and do not apply to D2887.
4. ASTM D86 distillation is run at Group 4 conditions except that Group 3 condenser temperatures are used.
5. The temperature difference between the temperature that demarks the 10 percent recovered point and the temperature that demarks the 50 percent recovered point must be at least 15 °C.
6. The temperature difference between the temperature that demarks the 10 percent recovered point and the temperature that demarks the 90 percent recovered point must be at least 40 °C.

3.5 Workmanship. The finished fuel shall be visually free from undissolved water, sediment, or suspended matter and shall be clear and bright. Refer to ASTM D6986. In case of dispute, the fuel shall be clear and bright at 21 °C (70 °F) and shall contain no more than 1.0 mg/L of particulate matter as required in Table IV.

3.6 Incidental contaminants. At the time of Government acceptance, the concentration of incidental contaminants in the finished fuel shall not exceed the limits in Table IV. For a definition of incidental contaminants, see 6.4.8.

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TABLE IV. Incidental contaminants

Material	Limits		ASTM or IP Test Method
	Minimum	Maximum	
Particulate matter (gravimetric), mg/L [1]		1.0	D2276 or D5452 [2] / IP 423
Filtration time, minutes [1]		15	
Particle counting, individual channel counts & ISO Code [3]	Channel Counts	ISO Code [4]	D8166 / IP 564, D7619 [2] / IP 565, or IP 577
≥ 4 µm(c) [5]	[6]	19	
≥ 6 µm(c) [5]	[6]	17	
≥ 14 µm(c) [5]	[6]	14	
≥ 30 µm(c) [5]	[6]	13	
Fatty Acid Methyl Ester (FAME) [7], mg/kg		50	D7797 / IP 583, IP 585 [2], IP 590, or IP 599

NOTES:

1. A minimum sample size of 3.785 liters (1 gallon) shall be filtered. Filtration time will be determined in accordance with procedure in Appendix A. This procedure may also be used for the determination of particulate matter as an alternate to ASTM D2276 or ASTM D5452.
2. Referee Test Method.
3. Alternate to the gravimetric particulate matter test method where equipment and laboratory capability exist. Should particulate counting result in failure, gravimetric particulate matter shall be the referee.
4. Per Table 1 "Allocation of scale numbers" of ISO 4406.
5. The notation (c) indicates that the equipment has been calibrated in accordance with ISO 11171.
6. Report.
7. FAME meeting the requirements of ASTM D6751 or EN 14214. For a definition of FAME, see 6.4.9.

3.7 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as quality conformance inspections (see 4.2).

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4.2 Qualification inspection conditions. Test for acceptance of individual lots shall consist of tests for all requirements specified in section 3. Quality conformance inspection shall include the test requirement herein.

4.2.1 Inspection lot. For acceptance purposes, individual lots shall be examined as specified herein and subjected to tests for all requirements cited in section 3.

4.3 Inspection.

4.3.1 Inspection conditions.

4.3.1.1 Refined hydrocarbon material. Fuel supplied from traditionally refined hydrocarbon distillate fuel oil meeting requirements of 3.1 shall comply with the specified limiting values in Table I using the cited test methods. The specified limiting values must not be changed. This precludes any allowance for test method precision and adding or subtracting digits. For the purposes of determining conformance with the specified limiting values, an observed value or a calculated value shall be rounded off “to the nearest unit” in the last right hand place of digits used in expressing the specified limiting value, in accordance with the Rounding Method of ASTM E29.

4.3.1.2 Synthesized hydrocarbon material. Fuel supplied containing synthesized materials meeting requirements of ASTM D7566 Annex A1 or Annex A2 as stipulated in 3.1.1 shall comply with the specified limiting values in Table I and Table III using the cited test methods. The specified limiting values must not be changed. This precludes any allowance for test method precision and adding or subtracting digits. For the purposes of determining conformance with the specified limiting values, an observed value or a calculated value shall be rounded off “to the nearest unit” in the last right hand place of digits used in expressing the specified limiting value, in accordance with the Rounding Method of ASTM E29.

4.4 Sampling plans.

4.4.1 Sampling. Each bulk or packaged lot of material shall be sampled for verification of product quality in accordance with ASTM D4057 or ASTM D4177, except where individual test procedures contain specific sampling instructions.

4.4.1.1 A number of jet fuel properties are very sensitive to trace contamination which can originate from sample containers. For recommended sample containers refer to ASTM D4306.

4.4.2 Sampling for inspection of filled containers. A random sample of filled containers shall be selected from each lot and shall be subjected to the examination of filled containers as specified in 4.5.1.3.

4.5 Methods of inspection.

4.5.1 Examination of product.

4.5.1.1 Visual inspection. Samples selected in accordance with 4.4.1 shall be visually examined for compliance with 3.5.

4.5.1.2 Examination of empty containers. Before filled, each unit container shall be visually inspected for cleanliness and suitability in accordance with ASTM D4057.

4.5.1.3 Examination of filled containers. Samples taken as specified in 4.4.2 shall be examined for conformance to MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing, and markings. Any container with one or more defects under the required fill shall be rejected.

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4.5.2 Chemical and physical tests. Tests to determine conformance to chemical and physical requirements of fuel supplied from traditionally refined hydrocarbon distillate fuel oil meeting requirements of 3.1 shall be conducted in accordance with Table I. The finished fuel shall pass all tests listed in Table I and Table III (should the finished fuel contain synthesized components). No additional testing shall be required. Requirements contained herein are not subject to corrections for test tolerances. If multiple determinations are made, results falling within any specified repeatability and reproducibility tolerances may be averaged. For rounding of significant figures, ASTM E29 shall apply to all tests required by this specification.

4.5.3 Thermal stability tests. The thermal stability test shall be conducted using ASTM D3241 at a temperature of 260 °C. Tube deposit ratings shall be measured using an Ellipsometric Tube Rater (D3241 Annex A3 ETR), when available; an Interferometric Tube Rater (D3241 Annex A2 ITR), when available; otherwise, a Visual Tube Rater (D3241 Annex A1 VTR). If test results by either ETR or ITR are reported, then results by the VTR are not required.

4.5.3.1 ASTM D3241 reported data. The following data shall be reported:

a. Differential pressure in millimeter of mercury (mm Hg) at 150 minutes, or time to differential pressure of 25 mm Hg, whichever comes first.

b. At the end of the test, the average deposit thickness in nanometers (nm) over an area of 2.5 mm^2 as determined with either the ETR or ITR, or the heater tube deposit rating as determined with the VTR.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The fuels covered by this specification are intended for use in aircraft turbine engines. JP-8 contains military-unique additives that are required by military weapon systems. This requirement is unique to military aircraft and engine designs. When authorized, JP-8 (NATO F-34) may be used in ground-based turbine and diesel engines. NATO F-35 is intended for commercial aviation, but can be converted to JP-8 (F-34) by the addition of the appropriate additives.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, date of this specification, and grade (type) of fuel.

b. Quantity required and size containers desired.

c. Level of packaging and packing required (see 5.1).

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d. Location and injection method for addition of electrical conductivity additive, fuel system icing inhibitor and corrosion inhibitor/lubricity improver, as required.

6.3 Conversion of metric units. Units of measure have been converted to the International System of Units (SI) (Metric) in accordance with IEEE/ASTM SI10 "American National Standard for Metric Practice." If test results are obtained in units other than metric or there is a requirement to report dual units, IEEE/ASTM SI10, should be used to convert the units.

6.4 Definitions.

6.4.1 Bulk lot. A bulk lot consists of an indefinite quantity of a homogeneous mixture of material offered for acceptance in a single isolated container or manufactured in a single plant run through the same processing equipment, with no change in ingredient material.

6.4.2 Packaged lot. A packaged lot consists of an indefinite number of 208-liter (55-gallon) drums, or smaller unit packages of identical size and type, offered for acceptance and filled from an isolated tank containing a homogeneous mixture of material; or filled with a homogeneous mixture of material run through the same processing equipment with no change in ingredient material.

6.4.3 Homogenous product. A homogeneous product is defined as a product where samples taken at various levels of the batch tank are tested for the defining homogeneous characteristics and all values obtained meet the repeatability precision requirements for that test method.

6.4.4 Synthesized Paraffinic Kerosene (SPK). Kerosene consisting of n-paraffins, iso-paraffins and cycloparaffins.

6.4.5 Fischer-Tropsch (FT) Process. A catalyzed chemical process in which carbon monoxide and hydrogen are converted into liquid hydrocarbons of various forms. Typical catalysts used are based on iron and cobalt.

6.4.6 Hydroprocessed Esters and Fatty Acids (HEFA) SPKs. Synthetic Paraffinic Kerosene produced by hydroprocessing plant, algal oils or animal fats.

6.4.7 Hydroprocessed or Hydrotreated Renewable Jet (HRJ). Terminology used to identify HEFA SPKs.

6.4.8 Incidental contaminants. These are known contaminants that can be picked up during fuel conveyance. These have defined upper allowable limits but are not identified as fuel additives. Examples include sediment (particulate) and FAME (biodiesel).

6.4.9 Fatty Acid Methyl Ester (FAME). FAME is synonymous with biodiesel meeting the requirements of ASTM D6751 or EN 14214. Per ASTM D6751, biodiesel is a "fuel comprised of long chain fatty acids derived from vegetable oils or animal fats, designated B100." JP-8 can pick up trace amounts of FAME when transported through conveyance methods, such as fungible product pipelines, where diesel fuel containing biodiesel is also transported.

6.5 Subject term (key word) listing.

Antioxidants
Biodiesel
Corrosion inhibitor
Fatty Acid Methyl Ester (FAME)
Fischer-Tropsch
Flash point
Freezing point

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Hydrocarbon distillate fuel
Hydrogen content
Hydroprocessed Esters and Fatty Acids (HEFA)
Hydroprocessed / Hydrotreated Renewable Jet (HRJ)
Icing inhibitor
Lubricity improver
Static dissipater
Synthesized Paraffinic Kerosene (SPK)
Thermal stability improver

6.6 International standardization agreement implementation. This specification implements:

a. NATO STANAG 1135 "Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations"
b. NATO STANAG/AFLP 3747 "Guide Specifications (Minimum Quality Standards) for Aviation Turbine Fuels (F-24, F-27, F-34, F-35, F-37, F-40 and F-44)"
c. AFIC AIR STD Fuels Group 4024 "Interchangeability Chart of Standardised Aviation Fuels, Lubricants, and Associated Products"

When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices, to change the agreement or make other appropriate accommodations.

6.7 Safety data sheets (SDS). Contracting officers will identify those activities requiring copies of completed SDS prepared in accordance with FED-STD-313 "Material Safety Data, Transportation Data, and Disposal Data for Hazardous Materials Furnished to Government Activities." The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.8 Test report. Test data required by 4.5 should be available for the procurement activity and user in the same order as listed in Table I. The Standard Form for Reporting Inspection Data on Aviation Turbine Fuels published in ASTM D1655 "Standard Specification for Aviation Turbine Fuels" should be used as a guide. Also, the type and amount of additives used should be reported.

6.9 Changes from previous issue. Marginal notes are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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METHOD FOR DETERMINATION OF FILTRATION TIME AND TOTAL SOLIDS

A.1 SCOPE

A.1.1 Scope. This Appendix describes the method for determining singularly or simultaneously the filterability characteristics and solids contamination of jet fuel. The purpose is to detect and prevent contaminants in jet fuel that can plug and cause rupture of ground filtration equipment, thereby affecting flight reliability of aircraft. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 METHOD

A.2.1 Summary of method. A 3.785 liter (1 gallon) volume of jet fuel is filtered through a membrane filter in the laboratory. The time required to filter this volume is measured in minutes and solids content is determined gravimetrically.

A.3 APPARATUS

a. Membrane filter: White, plain, 47 mm diameter, nominal pore size 0.8 µm. The membrane filter must conform to ASTM D5452 requirements.

b. Filtration apparatus: The apparatus, constructed of stainless steel, consists of a funnel and a funnel base with a filter support such that a membrane filter and a flow reducing washer can be securely held between the sealing surface of the funnel and funnel base (see Figure 1 "Apparatus for Determining Total Contaminant" in ASTM D5452).

c. Flow reducing washer: A 47-mm diameter flow reducer washer with an effective filtration area of 4.8 cm^2 (Millipore Corporation Part No. XX1004710, NSN 6640-01-018-2795).

d. Vacuum flask: A minimum of 4 liters.

e. Vacuum system: That develops in excess of 67.5 kPa (20 inches of mercury) vacuum.

f. Oven: Of the static type (without fan assisted circulation) controlling to 90 °C ± 5 °C (194 °F ± 9 °F).

g. Forceps: Flat-bladed with non-serrated non-pointed tips.

h. Dispenser, rinsing solvent (petroleum ether): Containing a 0.45 µm membrane filter in the delivery line. If solvent has been pre-filtered using a 0.45 µm filter then an inline filter is not required.

i. Glass petri dish: Approximately 125 mm in diameter with removable cover.

j. Analytical balance: Single or double pan, the precision standard deviation of which must be 0.07 mg or better.

A.4 PREPARATION

A.4.1 Preparation of apparatus and sample containers. All components of the filtration apparatus (except the vacuum flask), sample containers and caps must be cleaned as described in section 9 (Preparation of Apparatus and Sample Containers) of ASTM D5452. All metal parts of the filtration apparatus are to be electrically bonded and grounded, including the fuel sample container. See ASTM D5452 for other safety precautions.

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A.5 SAMPLING

A.5.1 Sampling. Obtain a representative 3.785 liter (1 gallon) sample as directed in section 8 (Sampling) of ASTM D5452. When sampling from a flowing stream is not possible, an all level sample or an average sample, in accordance with ASTM D4057 and/or ASTM D4177 shall be permitted. The 3.785 liter (1 gallon) sample container shall be an interior epoxy-coated metal can, a brown glass bottle, or a clear glass bottle protected by suitable means from exposure to light.

A.6 PROCEDURE

A.6.1 Test procedure.

a. Using forceps, place a new membrane (test) filter in a clean petri dish. Place the petri dish with the lid slightly ajar in a 90 °C ± 5 °C (194 °F ± 9 °F) oven for 30 minutes. Remove the petri dish from the oven and place it near the balance with the lid slightly ajar, but still protecting the filter from airborne contamination, for 30 minutes.

b. Weigh the test filter. A filter weighing in excess of 90 mg will not be used for time filtration testing.

c. Place a flow reducing washer (required only for time filtration testing) on top of funnel base then place a test filter on top of the reducing washer and secure the funnel to the funnel base.

d. Immediately prior to filtering the fuel, shake the sample to obtain a homogeneous mix and assure that fuel temperature does not exceed 30 °C (86 °F). Clean the exterior or top portion of the sample container to ensure that no contaminants are introduced. Any free water present in the fuel sample will invalidate the filtration time results by giving an excessive filtration time rating.

e. With the vacuum off, pour approximately 200 mL of fuel into the funnel.

f. Turn vacuum on and record starting time. Continue filtration of the 3.785 liters (1 gallon) sample, periodically shaking the sample container to maintain a homogenous mix. Record the vacuum (kPa or inches of mercury) 1 minute after start and again immediately prior to completion of filtration. Throughout filtration, maintain a sufficient quantity of fuel in the funnel so that the membrane filter is always covered.

g. Report the filtration time in minutes expressed to the nearest whole number. If filtration of the 3.785 liters (1 gallon) is not completed within 30 minutes, the test will be stopped and the volume of the fuel filtered will be measured. In these cases, report filtration time as ">30 minutes" and the total volume of fuel filtered.

h. Report the vacuum (kPa or inches of mercury) as determined from the average of the two readings taken in A.6.1.f.

i. After recording the filtration time, shut off the vacuum and rinse the sample container with approximately 100 mL of filtered petroleum ether and dispense into the filtration funnel. Turn vacuum on and filter the 100 mL rinse. Turn vacuum off and wash the inside of the funnel with approximately 50 mL of filtered petroleum ether. Turn vacuum on and filter. Repeat the funnel rinse with another 50 mL of petroleum ether but allow the rinse to soak the filter for approximately 30 seconds before turning the vacuum on to filter the rinse. With vacuum on, carefully remove the top funnel and rinse the periphery of the filter by directing a gentle stream of petroleum ether from the solvent dispenser from the edge of the filter toward the center,

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taking care not to wash contaminants off the filter. Maintain vacuum after final rinse for a few seconds to remove the excess petroleum ether from the filter.

j. Using forceps, carefully remove test filter (from the funnel base and flow reducing washer if present) and place in a clean petri dish. Dry in the oven at 90 °C ± 5 °C (194 °F ± 9 °F) for 30 minutes with the cover on the petri dish slightly ajar. Remove the petri dish from the oven and place it near the balance with the lid slightly ajar, but still protecting the filter from airborne contamination, for 30 minutes. Reweigh the filter.

k. Report the total solids content in mg/L by using the following formula:

$$\frac{\text{weight gain of filter in mg}}{3.785\ \text{L}} = \text{mg/L}$$

l. Should the sample exceed the 30-minute filtration time and a portion of the fuel is not filtered, the solids content in mg/liter will be figured as follows: Determine the volume of fuel filtered by subtracting the mL of fuel remaining from 3785 mL.

$$\frac{\text{weight gain of filter in mg}}{\text{mL of fuel filtered} \times 0.001} = \text{mg/L}$$

A.6.2 Test conditions for filtration time.

a. The vacuum should exceed 67.5 kPa (20 inches of mercury) throughout the test. The differential pressure across the filter should exceed 67.5 kPa (20 inches of mercury).

b. The fuel temperature shall be between 18 °C and 30 °C (64 °F and 86 °F). If artificial heat (such as a hot water bath) is used to heat the sample, erroneously high filtration times may occur, but this approach is allowed.

A.7 NOTES

A.7.1 Filtration time. If it is desired to determine the filtration time and not the total solids content, perform the test by omitting steps A.6.1.i, A.6.1.j, A.6.1.k, and A.6.1.l.

A.7.2 Total solids. If it is desired to determine the total solids content and not the filtration time, use of the flow reducing washer may be omitted. When a reducing ring is not used, then total solids shall be determined per ASTM D5452 and the use of a control filter shall be required.

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CONCLUDING MATERIAL

Custodians:
Navy – AS
Army – AT
Air Force – 68
DLA – PS

Review activities:
Army – AR, AV
Air Force – 11, 20, 184

Preparing activity:
Air Force – 68
(Project 9130-2018-002)

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